Hierarchical Approach for Extracting HPO Codes from Clinical Notes

[1]Brooke Ballhaus [2]Riya Narain

[1]Department of Computer Science, Stanford University [2]Department of Bioengineering, Stanford University

Project Overview

In modern clinical practice, a key challenge in rare disease diagnosis is accurately identifying phenotypic abnormalities, which the Human Phenotype Ontology (HPO) represents using standardized codes [3]. However, manual annotation of clinical notes with HPO codes is time-consuming and prone to error, limiting diagnostic accuracy and large-scale research.

While models like PhenoBERT [2] and DeepTag [5] have shown promise, they struggle with complex hierarchical relationships and context-dependent interpretations of clinical terms.

To address these limitations, we introduce the **HierTag** model, a hybrid approach combining the hierarchical learning of DeepTag with PhenoBERT's text segmentation. Key features include:

- A hierarchical training objective to improve alignment with HPO structure.
- Modified loss functions to penalize misclassifications based on semantic distance.
- An abstention mechanism to reduce false positives by deferring uncertain predictions.

HierTag enhances the accuracy and reliability of HPO coding, offering a scalable solution for rare disease diagnosis and phenotype-driven research.

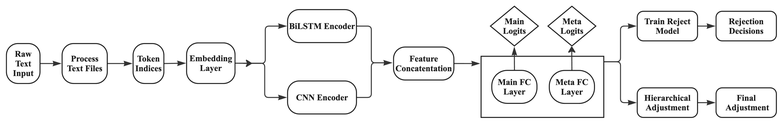


Figure 1. HierTag Model Architecture.

Dataset and Metrics

Datasets:

- **GSC+** [4]: 228 biomedical literature abstracts encompassing diverse clinical language.
- **ID-68** [1]: 68 clinical descriptions focused on intellectual disabilities, presenting real-world challenges.

Example Input/Output:

Input	Output	
An 8 years old girl who was born at full term via NSVD to a 33-year-old mother. A decrease in fetal movements was noted during pregnancy, and her birth weight was 2.8 kg. After birth, she was noted to be floppy with poor sucking. At one month of age, she began having choking episodes due to severe GERD, requiring a Niessel fundoplication. Aspiration problems persisted, and the procedure was repeated. She also developed seizures, which continue despite multiple medications.	decrease in fetal movements: with poor sucking: choking episodes: GERD: Aspiration problems: seizures:	HP:0001558 HP:0002033 HP:0030842 HP:0002020 HP:0002835 HP:0001250

Table 1. Example of Input and Output Data from the ID-68 Dataset.

Evaluation Metrics: We assess model performance using precision, recall, and F1-score. These metrics are computed as both micro- and macro-averages to capture performance across common and rare phenotypic classes.

Methods and Experiments

Baseline (PhenoBERT): Clinical texts are segmented into clinically relevant text segments (CTSs). PhenoBERT applies a CNN-based candidate selection, followed by a BERT encoder to produce initial HPO code predictions.

HierTag Model: Our proposed model extends this approach with the following components:

- **CNN Encoder:** Extracts local features using the equation:

$$h_i = \mathrm{ReLU}(W_{\text{conv}} * x_i + b)$$

  A max pooling operation aggregates these features into a fixed-length representation:

$$x = [\max(h_1), \max(h_2), \max(h_3)]$$

- **BLSTM for Hierarchical Learning:** To capture long-range dependencies, a Bidirectional LSTM processes the sequence:

$$h_f^t = \mathrm{LSTM}_f(x_t,\ h_f^{t-1}), \quad h_b^t = \mathrm{LSTM}_b(x_t,\ h_b^{t+1})$$

  Max pooling over these outputs produces a unified representation of the input sequence.
- **Loss Functions:** The model is trained using a composite loss function that includes:
  - **Binary Cross-Entropy (BCE) Loss:** To drive the model to predict individual HPO codes accurately, we compute the following loss using BCE:

$$L_{BCE} = -\sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

  - **Meta Loss:** To encourage learning of higher-level, aggregated patterns among phenotypes and to guide the model to generalize better across related phenotypic groups, meta loss is computed using binary cross-entropy on meta labels:

$$L_{\text{meta}} = -\sum_i [y_{\text{meta},i} \log(p_{\text{meta},i}) + (1 - y_{\text{meta},i}) \log(1 - p_{\text{meta},i})]$$

  - **Co-occurrence Loss:** To capture relationships between phenotype pairs, the co-occurrence loss penalizes deviations between predicted and observed co-occurrence frequencies:

$$L_{\text{co-occ}} = \sum_{i,j} f(X_{ij}) \left(w_i \cdot w_j + b_i + b_j - \log(X_{ij})\right)^2$$

  - **Abstention Loss:** An additional loss component penalizes discrepancies between the reject model's output R_i and the per-sample loss L_i:

$$L_{\text{abst}} = \frac{1}{N} \sum_{i=1}^{N} (R_i - L_i)^2$$

- **Abstention Mechanism:** A reject model assesses the confidence of each prediction and defers decisions for low-confidence cases, effectively reducing false positives.
- **Training Details:** The model is implemented in PyTorch using BioBERT embeddings and trained with the Adam optimizer (learning rate = 0.001) for 65 epochs with a batch size of 32.

Discussions and Future Research

Discussions: Our experiments indicate that the HierTag model consistently outperforms the PhenoBERT baseline, particularly in terms of precision and F1-score. Ablation studies show that the abstention mechanism is critical: without it, precision drops by approximately 3.4% even if recall increases slightly. These results underscore the importance of modeling hierarchical dependencies and incorporating uncertainty handling in clinical decision-making.

Future Research: Future work will address several challenges:

- **Enhancing Rare Phenotype Detection:** Incorporating targeted data augmentation and fine-tuning strategies to better capture underrepresented classes.
- **Refining Hierarchical Representations:** Exploring advanced architectures and loss formulations to more precisely model subtle differences among closely related phenotypes.
- **Integrating Multimodal Data:** Expanding the framework to include additional clinical modalities such as imaging and genetic data.
- **Improving Interpretability:** Developing visualization and explanation tools to help clinicians understand and trust model predictions.

Results

The performance of the HierTag model was evaluated on both the GSC+ and ID-68 datasets. The tables below summarize precision, recall, and F1-scores computed as both micro- and macro-averages.

GSC+ Test Set:

Method	Micro Average (%)			Macro Average (%)		
	P	R	F1	P	R	F1
PhenoBERT	79.25	65.97	72.00	79.11	70.64	74.64
HierTag Model	**80.15**	**71.84**	**75.77**	**81.76**	70.05	**75.45**

Table 2. Performance on the GSC+ Test Set.

ID-68 Test Set:

Method	Micro Average (%)			Macro Average (%)		
	P	R	F1	P	R	F1
PhenoBERT	93.89	**81.57**	87.30	94.09	**79.02**	85.89
HierTag Model	**96.51**	79.94	**87.45**	**96.42**	77.73	**86.07**

Table 3. Performance on the ID-68 Test Set.

Interpretation:

- **Improved Precision:** The HierTag model shows a clear advantage in precision. This improvement is primarily due to the integrated abstention mechanism, which allows the model to defer predictions when confidence is low, thereby reducing the incidence of false positives.
- **F1-Score Gains:** The F1-score improvements on both datasets indicate that HierTag achieves a better balance between precision and recall. On the GSC+ dataset, the F1-score improves by approximately 3.77 points.
- **Macro-Average Considerations:** Although HierTag improves overall performance, the macro-average metrics remain lower relative to micro-average scores. This suggests that while the model performs well on high-support, frequently occurring phenotypes, it struggles more with rare classes. The abstention mechanism contributes to this by deferring predictions on uncertain (often rare) cases, which, while reducing false positives, can also lead to a drop in recall for those classes.
- **Recall Trade-offs:** On the ID-68 dataset, PhenoBERT shows a slightly higher recall, indicating it predicts more of the positive cases overall. However, the HierTag model's higher precision results in a more reliable overall performance, as reflected in its improved F1-score.

Overall, these results confirm that our hierarchical approach combined with an effective abstention strategy significantly enhances the reliability of HPO code extraction, particularly in reducing misclassifications in common phenotypic classes, while still highlighting challenges in rare phenotype detection.

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